ORBITL1-AX25

PHASE 3 TECHNICAL REPORT | END-TO-END INTEGRATION & INDEPENDENT INTEROPERABILITY

Phase 3 COMPLETE	22 / 22 Tests PASS	100% Bit-Accurate	ISO/IEC & AX.25 v2.2
E2E Integration Gate Status	E2E & Interop Suite	Reference Model Agreement	Authoritative Vector Match

Phase 3 Architectural Scope & Objectives

Phase 3 validates complete end-to-end integration and external interoperability for the ORBITL-1 AX.25 communication system. It integrates the Layer 2 packet builder (Phase 1) with the bit-framing and NRZI line coding engine (Phase 2) into a cohesive pipeline. To establish rigorous compliance without vendor bias, Phase 3 cross-validates the software against (1) authoritative ISO/IEC 3309 CRC catalog vectors, (2) published AX.25 v2.2 specification address vectors (Section 3.12), and (3) an independent, from-scratch pure-Python reference model.

Phase 3 Automated Test Suite Composition (tests/test_integration.py)

Test Class	Test Count	Functional Focus	Key Assertions & Verification Conditions	Pass Rate
TestEndToEndIntegration	9	Full TX -> RX Lifecycle	Arbitrary chunk sizes (1 to 31 bits), boundary sizes (4078B), shared 0x7E delimiter, consecutive frames	100% (9/9)
TestErrorRecoveryAndRejection	8	Fault Injection & Robustness	Single/double bit flips in FCS, HDLC abort sequence (7+ ones), oversized frames, line noise resync	100% (8/8)
TestIndependentInteroperability	5	Authoritative Vector Conformance	ISO/IEC 3309 CRC vectors, AX.25 v2.2 Sec 3.12 address vectors, FCS residual constant, Reference Model	100% (5/5)
TOTAL PHASE 3	22	Comprehensive Integration Suite	Verified end-to-end data preservation and independent standards interoperability	100% (22/22)

Phase 3 Interoperability Evidence & Independent Validation

Bit-for-bit cross-validation against authoritative international standards and reference models.

1. Authoritative Specification Vector Verification Matrix

Standard Specification	Test Vector / Input Vector	Expected Specification Value	Actual System Value	Verification Result
ISO/IEC 3309 / ITU-T X.25	Standard ASCII: '123456789'	CRC-16 = 0x906E	0x906E	VERIFIED (100% Match)
ISO/IEC 3309 / ITU-T X.25	Null Payload: b''	CRC-16 = 0x0000	0x0000	VERIFIED (100% Match)
ISO/IEC 3309 / ITU-T X.25	Single Byte: b'A'	CRC-16 = 0xA3F5	0xA3F5	VERIFIED (100% Match)
AX.25 v2.2 (Section 3.12)	Destination 'NJ7P', SSID 0 (Not Last)	9C 94 6E A0 40 40 60	9C 94 6E A0 40 40 60	VERIFIED (100% Match)
AX.25 v2.2 (Section 3.12)	Source 'N7LEM', SSID 0 (Last Address)	9C 6E 98 8A 9A 40 61	9C 6E 98 8A 9A 40 61	VERIFIED (100% Match)
HDLC Mathematical Residue	FCS Residual Constant	Residual Constant = 0xF0B8	0xF0B8 (Residue Verified)	VERIFIED (100% Match)

2. Pure-Python Independent Reference Model (IndependentAX25Reference)

To guarantee that the implementation contains no circular logic or shared implementation biases, an entirely separate, independent reference model was authored in tests/test_integration.py. The reference model independently computes bitwise HDLC stuffing, LSB-first bit serialization, and NRZI transitions. Both nominal and pathological datasets were passed through both implementations, confirming 100% bit-for-bit identity across every bit position.

Dataset Tested	Data Size	Reference Model Output	Production Pipeline Output	Bit Agreement
Random Telemetry Payloads	16, 64, 128, 256 bytes	Bitstream generated independently	Bitstream generated via pipeline	100.00% (Identical)
Dense Ones (0xFF Stress)	50 contiguous 0xFF bytes	Bit-stuffing zero inserted every 5 ones	Exact identical bit positions	100.00% (Identical)
Embedded Flag Octets (0x7E)	25 contiguous 0x7E bytes	Bit-stuffing breaks flag pattern	Exact identical bit positions	100.00% (Identical)

Phase 3 Streaming Stress & Fault Injection Recovery

System resilience under arbitrary chunk sizes, line noise, and deliberate byte corruptions.

1. Prime-Sized & Arbitrary Streaming Chunk Matrix

Chunk Size Tested	Simulation Purpose	NRZI State Retention	Frame Recovery Outcome
1 bit (Single bit)	Simulates finest-grain bit-by-bit hardware shift register	Retains level state across all 8,000+ chunk transitions	PASS (100% Packet Match)
2, 3, 5, 7 bits (Prime sizes)	Prevents accidental alignment with byte or octet boundaries	Shift registers advance correctly across unaligned boundaries	PASS (100% Packet Match)
11, 13, 17, 31 bits	Simulates variable UART/FIFO buffer read interrupts	Zero buffer drift or missed flag sequence	PASS (100% Packet Match)
128 bits (16 bytes)	Simulates bulk DMA block transfers	High-throughput block parsing without latency	PASS (100% Packet Match)

2. Fault Injection & Rejection Verification Matrix

Injected Fault Condition	Test Method	Detection Mechanism	Observed System Behavior
Single-Bit FCS Flip	Inverted 1 bit in 16-bit CRC field	crc16_x25() recalculation mismatch	Corrupted frame dropped cleanly; 0 corrupt bytes delivered
Double-Bit FCS Flip	Inverted 2 bits in 16-bit CRC field	crc16_x25() recalculation mismatch	Corrupted frame dropped cleanly; 0 corrupt bytes delivered
HDLC Abort Sequence	Injected 7 contiguous 1-bits (0x7F)	HDLC state machine abort handler	Active frame discarded; receiver waits for next 0x7E flag
Line Noise Bursts	Injected random bytes before valid frame	Hunting state discards noise octets	Automatically synchronizes upon valid opening 0x7E delimiter

Phase 3 Source File Integrity & Software Acceptance

Cryptographic baseline verification, hardware readiness, and formal sign-off.

Cryptographic Production Source Hash Matrix (SHA-256)

Source Code Module	File Path	SHA-256 Checksum	Preservation Status
AX.25 Packet Protocol	src/ax25.py	8da3d41a47c8211e482270a554cbb04e046867159dadfb789e8944990e30b9f1	VERIFIED (100% Unmodified)
NRZI & HDLC Primitives	src/ax25_hdlc.py	38fbab7afe40a822bbee0ea0b0c5047deede1ab1f05306813e01ff11a4894391	VERIFIED (100% Unmodified)
Streaming Receive Pipeline	src/ax25_receiver.py	7d9b6c1df0af7636f4babdf1e9a980f425045ee63a17c2740f0f3c681bb87d68	VERIFIED (100% Unmodified)
AX.25 Transmitter Pipeline	src/ax25_transmitter.py	6120408159812b97dfc1612a54c8f58d54bd76c4810d9941ce29280c2029aca8	VERIFIED (100% Unmodified)
Mobilinkd HDLC Engine	src/vendor/mobilinkd/HDLC.py	db5c61e16cbff27cce26d244ed71afffdb4c42c036588c57c990e6754dbe7b36	VERIFIED (100% Unmodified)

Phase 3 Integration Readiness Assessment

With Phase 3 complete, the full bidirectional software pipeline is verified. Incoming and outgoing frames conform exactly to amateur satellite standards. The software is fully isolated, tested with 22 rigorous integration tests, and ready for Phase 4 stress benchmarks.

Formal Software Acceptance Gate Declaration

PHASE 3 ACCEPTANCE GATE: MET (100% SOFTWARE-VERIFIED)
All 22 automated integration and interoperability tests in tests/test_integration.py execute cleanly.
Bit-for-bit agreement with the Independent Reference Model and international standards (ISO/IEC 3309 and AX.25 v2.2) is established.
Zero regressions were observed across Phase 1 and Phase 2 baselines.